

Threat Detection & SIEM Pipeline

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Overview

This project demonstrates an end-to-end cybersecurity threat detection pipeline using Kali Linux (Attacker), Windows 11 (Target Endpoint), and Splunk Enterprise (SIEM). By simulating authentication attempts and network reconnaissance against SMB services, this lab captures raw endpoint telemetry, validates SIEM log ingestion, and correlates Windows Security events for threat detection.

Architecture

Component	OS/Role	IP address	Purpose
Attacker	Kali Linux	10.0.0.20	Network Scanning and SMB Authentication testing.
Target Endpoint	Windows 11 Pro	10.0.0.10	Host SMB service and running Splunk Universal Forwarder.
SIEM/Indexer	Ubuntu Server / Splunk Enterprise	10.0.0.30	Centralized log ingestion, indexing, and alert analysis.

Implementation

Phase 1: Reconnaissance & Ingestion Baseline

1. Host Discovery & Port Scanning

- Confirmed host reachability via ICMP ping and executed Nmap service detection scan (nmap -sV -sC -O) against target host 10.0.0.10, revealing 1,000 filtered ports managed by host firewall rules.

```
homelab@HomeLab: ~  
Session Actions Edit View Help  
  
(homelab@HomeLab)~  
$ ping -c 3 10.0.0.10  
PING 10.0.0.10 (10.0.0.10) 56(84) bytes of data.  
64 bytes from 10.0.0.10: icmp_seq=1 ttl=128 time=1.62 ms  
64 bytes from 10.0.0.10: icmp_seq=2 ttl=128 time=0.800 ms  
64 bytes from 10.0.0.10: icmp_seq=3 ttl=128 time=0.589 ms  
  
— 10.0.0.10 ping statistics —  
3 packets transmitted, 3 received, 0% packet loss, time 2010ms  
rtt min/avg/max/mdev = 0.589/1.003/1.620/0.444 ms  
  
(homelab@HomeLab)~  
$ nmap -sV -sC -O 10.0.0.10  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-26 17:42 -0400  
Nmap scan report for 10.0.0.10  
Host is up (0.00070s latency).  
All 1000 scanned ports on 10.0.0.10 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
MAC Address: 08:00:27:49:CD:9D (Oracle VirtualBox virtual NIC)  
Too many fingerprints match this host to give specific OS details  
Network Distance: 1 hop  
  
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 34.79 seconds  
  
(homelab@HomeLab)~  
$
```

2. SIEM Telemetry Ingestion Baseline

- Splunk Enterprise search results (index="main") confirming active log ingestion from Windows 11 endpoint (admin-samuel) with over 38,000 security telemetry events captured.

The screenshot displays the Splunk Enterprise Search interface. At the top, the navigation bar includes 'splunk>enterprise', 'Apps', 'Administrator', 'Messages', 'Settings', 'Activity', 'Help', and a search bar. Below this, the 'Search & Reporting' section is active. The 'New Search' panel shows the search query 'index="main"' with a search button. Below the query, it indicates '38,866 events (before 7/26/26 9:44:31.000 PM)' and 'No Event Sampling'. The interface includes tabs for 'Events (38,866)', 'Patterns', 'Statistics', and 'Visualization'. The 'Events' tab is selected, showing a timeline visualization and a list of events. The list view shows a table with columns for 'Time' and 'Event'. The first event is from 7/27/26 at 12:44:06.780 AM, with details including LogName=Security, EventCode=5379, EventType=0, ComputerName=admin-samuel, and Show all 21 lines. The bottom left shows 'SELECTED FIELDS' (host 1, source 3, sourcetype 3) and 'INTERESTING FIELDS' (Account_Domain 9, Account_Name 19). A 'Show desktop' button is visible in the bottom right corner.

Time	Event
7/27/26 12:44:06.780 AM	07/26/2026 05:44:06.780 PM LogName=Security EventCode=5379 EventType=0 ComputerName=admin-samuel Show all 21 lines host = admin-samuel source = WinEventLog:Security sourcetype = WinEventLog:Security

- Parsed field extractions for Event Code 5379 on host admin-samuel, demonstrating active log normalization, indexing, and telemetry capture within Splunk Enterprise.

Time	Event	1	2	3	4	5	6	7	8	Next >
	EventCode=5379 EventType=0 ComputerName=admin-samuel Show all 21 lines									
	Event Actions ▾									
Type	☒ Field	Value	Actions							
Selected	☒ host ▾	admin-samuel	▾							
	☒ source ▾	WinEventLog:Security	▾							
	☒ sourcetype ▾	WinEventLog:Security	▾							
Event	☐ Account_Domain ▾	WORKGROUP	▾							
	☐ Account_Name ▾	ADMIN-SAMUEL\$	▾							
	☐ ComputerName ▾	admin-samuel	▾							
	☐ EventCode ▾	5379	▾							
	☐ EventType ▾	0	▾							
	☐ Keywords ▾	Audit Success	▾							
	☐ LogName ▾	Security	▾							
	☐ Logon_ID ▾	0x3E7	▾							
	☐ Message ▾	Credential Manager credentials were read. Subject: Security ID: S-1-5-18 Account Name: ADMIN-SAMUEL\$ Account Domain: WORKGROUP Logon ID: 0x3E7 Read Operation: Enumerate Credential	▾							

Phase 2: Threat Simulation & Endpoint Logging

1. Automated Credential Testing
 - Credential Access Phase: Executed SMB brute-force attack using Hydra against target 10.0.0.10 over port 445 to trigger target host logging and authentication failure telemetry.

```
(homelab@HomeLab)-[~]
$ hydra -l admin -P /usr/share/wordlists/fasttrack.txt 10.0.0.10 smb
Hydra v9.7 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-07-26 17:56:03
[INFO] Reduced number of tasks to 1 (smb does not like parallel connections)
[DATA] max 1 task per 1 server, overall 1 task, 262 login tries (l:1/p:262), ~262 tries per task
[DATA] attacking smb://10.0.0.10:445/
```

2. Direct Authentication Testing
 - Authentication Testing Phase: Executed direct SMB connection attempt against the IPC\$ share on target 10.0.0.10 using smbclient, generating an NT_STATUS_LOGON_FAILURE to test endpoint logging controls.

```
homelab@HomeLab: ~
Session Actions Edit View Help

(homelab@HomeLab)-[~]
$ smbclient //10.0.0.10/IPC$ -U testuser
Password for [WORKGROUP\testuser]:
session setup failed: NT_STATUS_LOGON_FAILURE

(homelab@HomeLab)-[~]
$
```

Phase 3: SIEM Correlated Detection Analysis

1. Event Code 4625 Verification
- Telemetry Capture Phase: Detailed Splunk event payload of Windows Event Code 4625 (Logon Type 3 - Network) confirming authentication failure logging on host admin-samuel for targeted user 'testuser'.

i	Time	Event
>	7/27/26 1:28:35.161 AM	07/26/2026 06:28:35.161 PM LogName=Security EventCode=4625 EventType=0 ComputerName=admin-samuel SourceName=Microsoft Windows security auditing. Type=Information RecordNumber=59559 Keywords=Audit Failure TaskCategory=Logoff OpCode=Info Message=An account failed to log on. Subject: Security ID: S-1-0-0 Account Name: - Account Domain: - Logon ID: 0x0 Logon Type: 3 Account For Which Logon Failed: Security ID: S-1-0-0 Account Name: testuser Account Domain: WORKGROUP

Time	Event
	Security ID: S-1-0-0 Account Name: testuser Account Domain: WORKGROUP Failure Information: Failure Reason: Unknown user name or bad password. Status: 0xC000006D Sub Status: 0xC0000064 Process Information: Caller Process ID: 0x0 Caller Process Name: - Network Information: Workstation Name: HOMELAB Source Network Address: 10.0.0.20 Source Port: 39066 Detailed Authentication Information: Logon Process: NtLmSsp Authentication Package: NTLM Transited Services: - Package Name (NTLM only): - Key Length: 0

This event is generated when a logon request fails. It is generated on the computer where access was attempted.

Lessons Learned & Mitigation Controls

1. Enabling Failure Auditing via Policy

- Ensure logon failure auditing is explicitly enforced across Windows endpoints so telemetry is generated for the SIEM forwarder:

PowerShell

<code>auditpol /set /subcategory:"Logon" /failure:enable /success:enable</code>

2. Configure Account Lockout Policies

- Implement Local Policy thresholds to lock user accounts after a set number of consecutive failed attempts (e.g., 5 attempts within 15 minutes) to neutralize automated dictionary attacks.

3. Restrict SMB Exposure via Network Firewalls

- Block inbound SMB (TCP Port 445) access from untrusted segments, allowing connection requests strictly from designated management or administrative subnets.

4. Enforce Strong Authentication Policies

- Implement strong password complexity enforcement and Multi-Factor Authentication (MFA) across remote network services to prevent unauthorized account access.